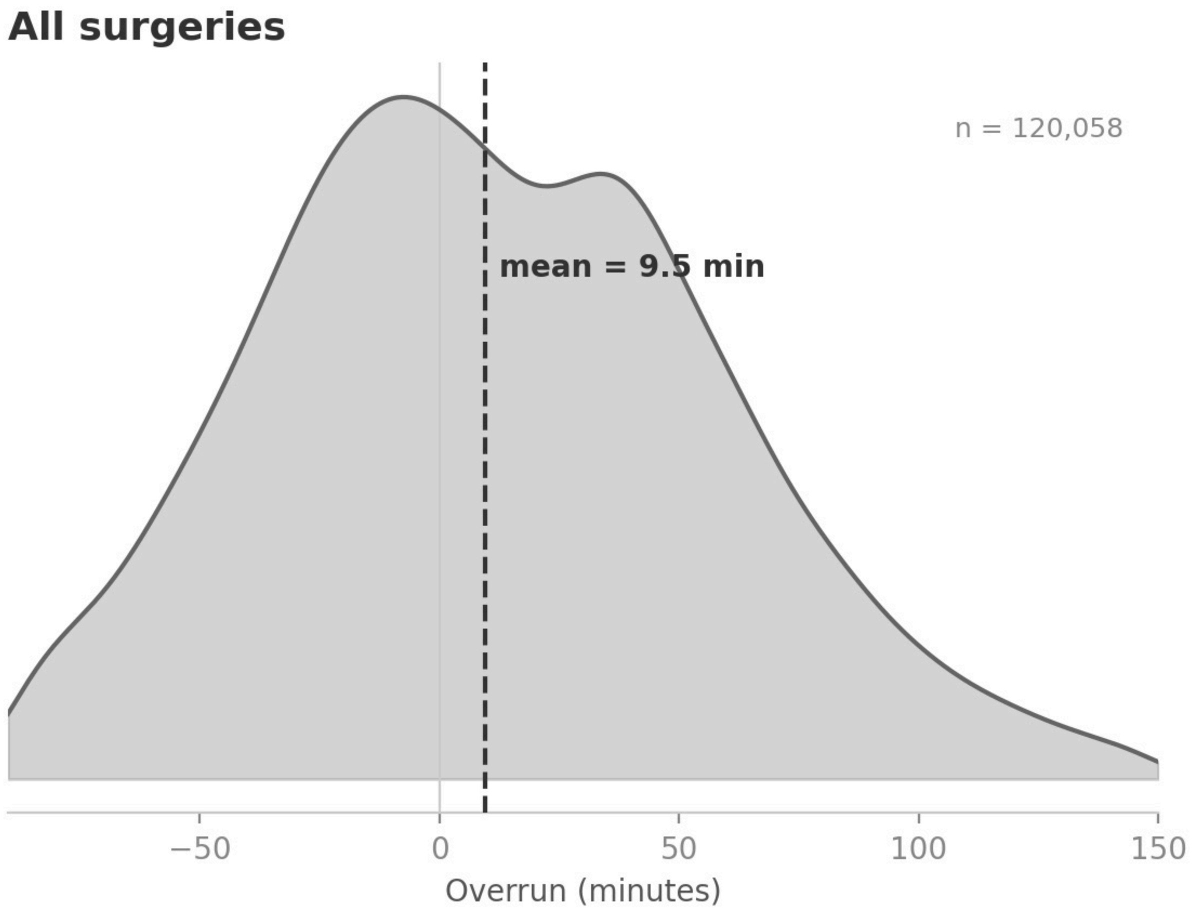
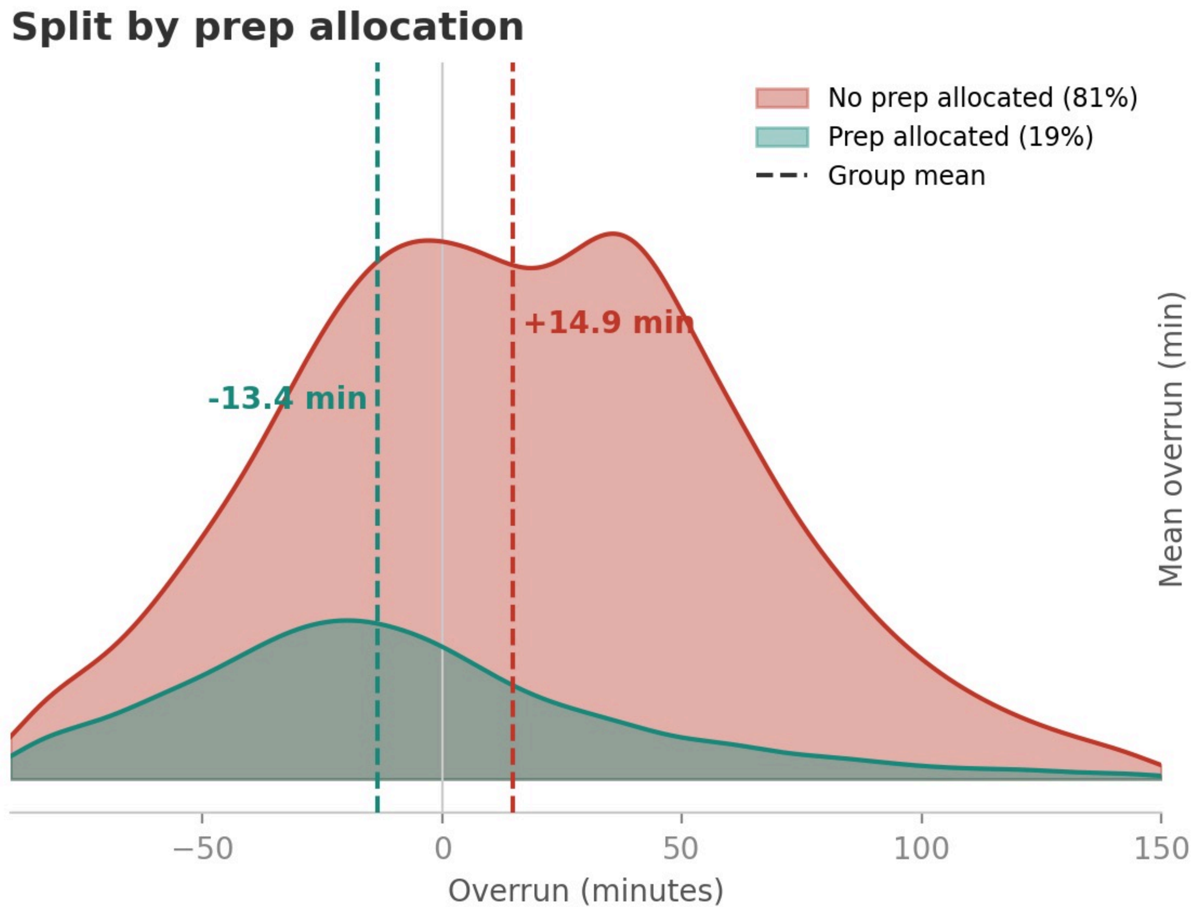
Why Surgeries Run Late: A Scheduling Fix for Rigshospitalet

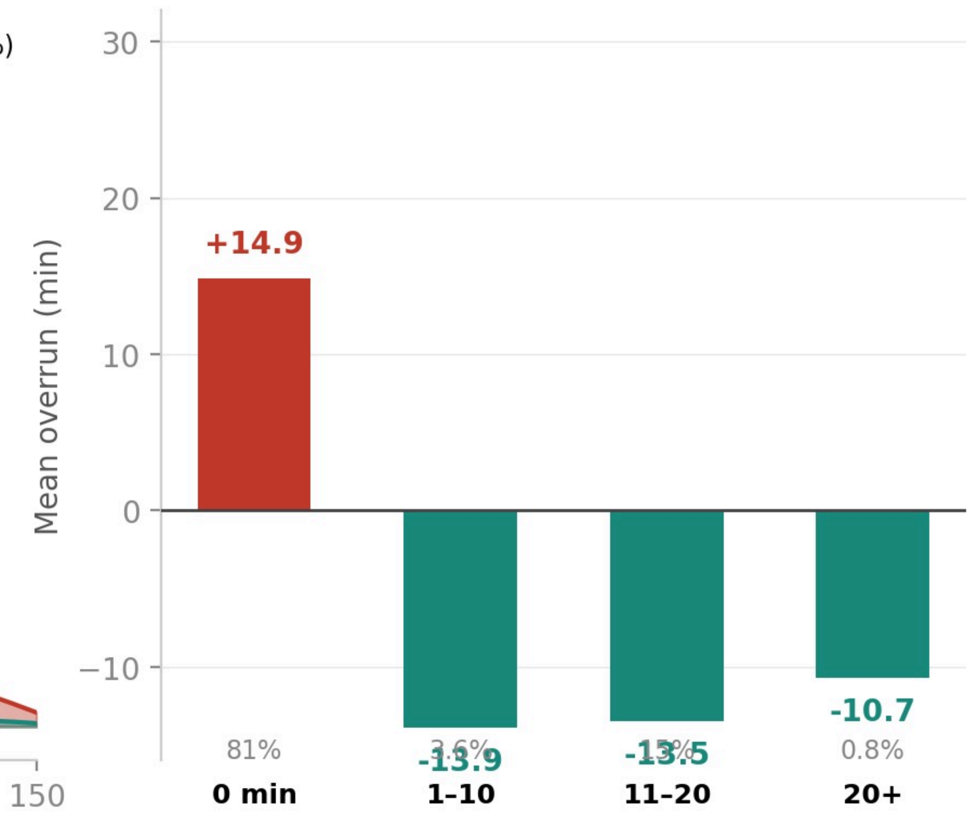
120,868 surgeries · 32 specialties · 2 years

Authors: Linus Juni, Theodor la Cour, Mathilde Sørensen, Jose Diaz, Rune Isaksen & Alexander Sucur

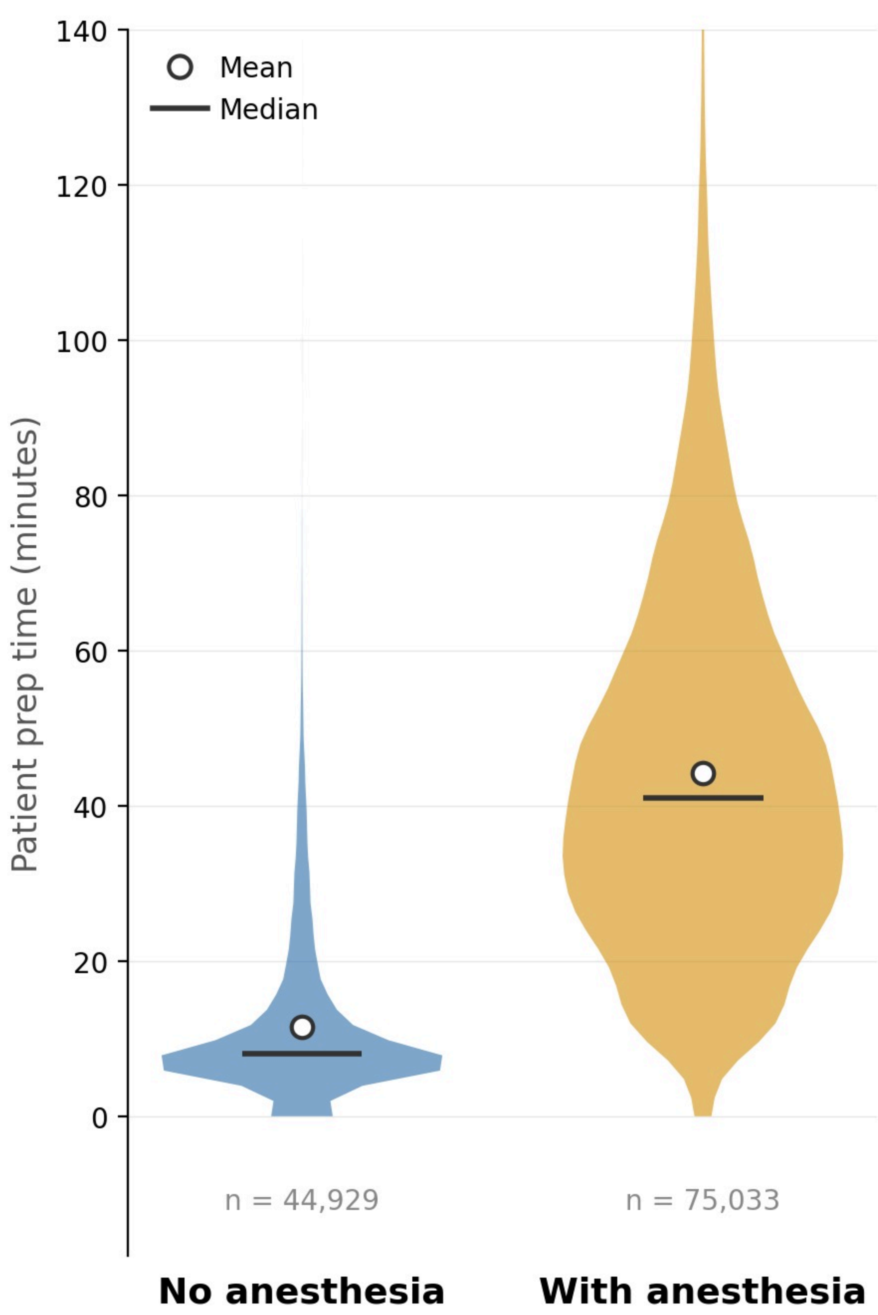
Problem: The schedule allocates zero prep time for 81% of surgeries

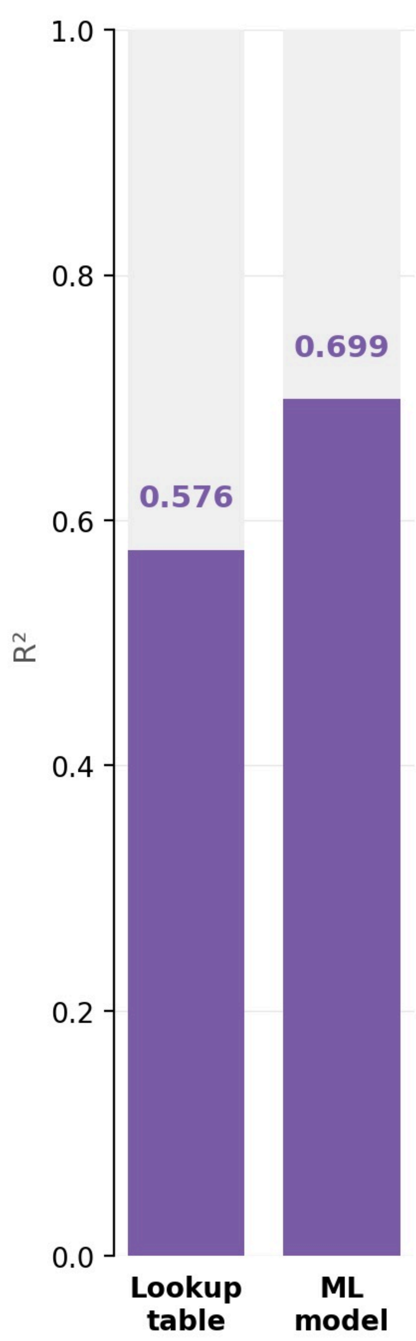

All surgeries
n = 120,058
mean = 9.5 min
−50
0
50
100
150
Overrun (minutes)

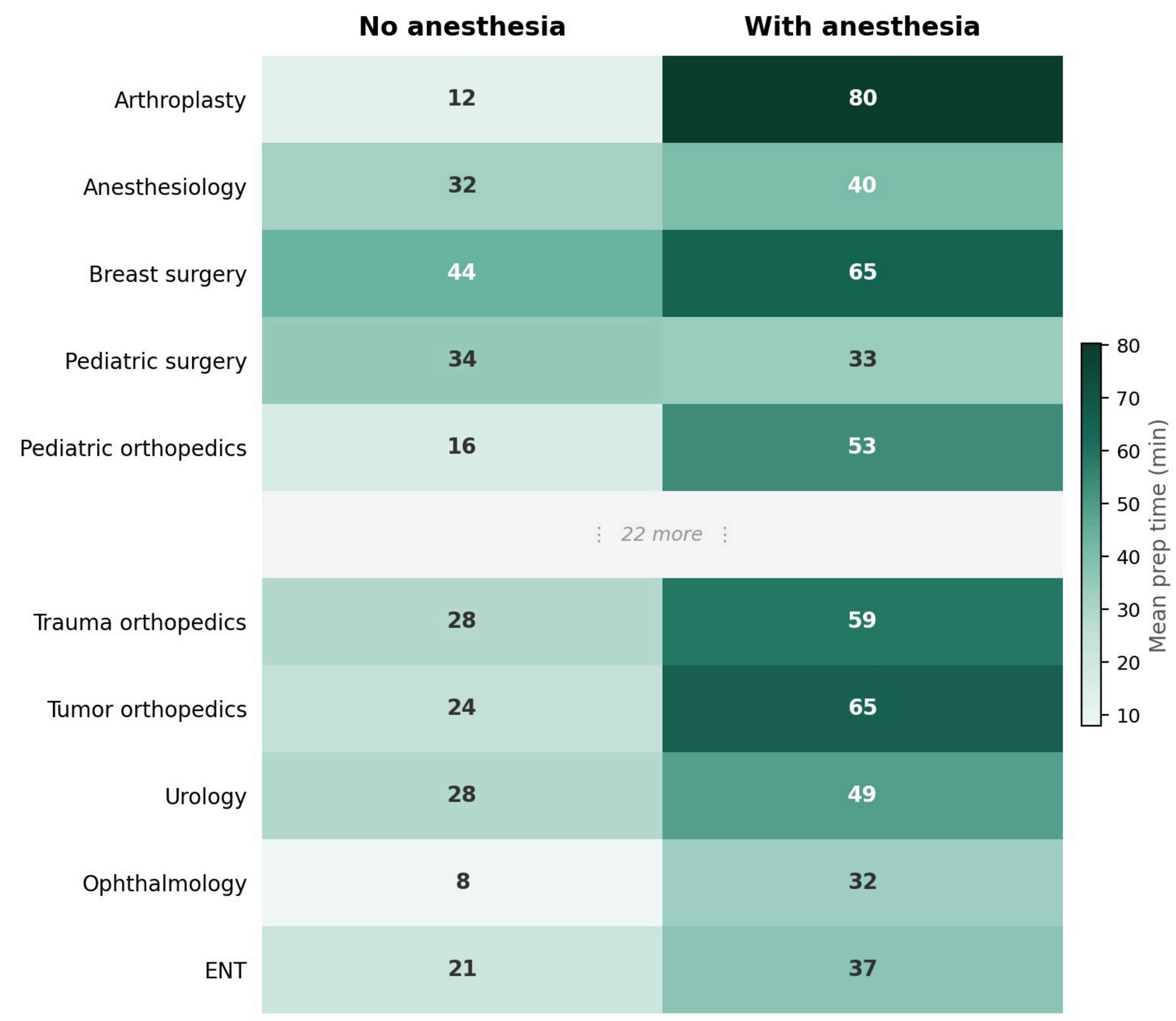

Split by prep allocation
No prep allocated (81%)
Prep allocated (19%)
Group mean
-13.4 min
+14.9 min
−50
0
50
100
150
Overrun (minutes)


By prep bucket
Mean overrun (min)
30
20
10
0
−10
+14.9
-13.9
-13.5
-10.7
81%
3.6%
0.8%
0 min
1–10
11–20
20+

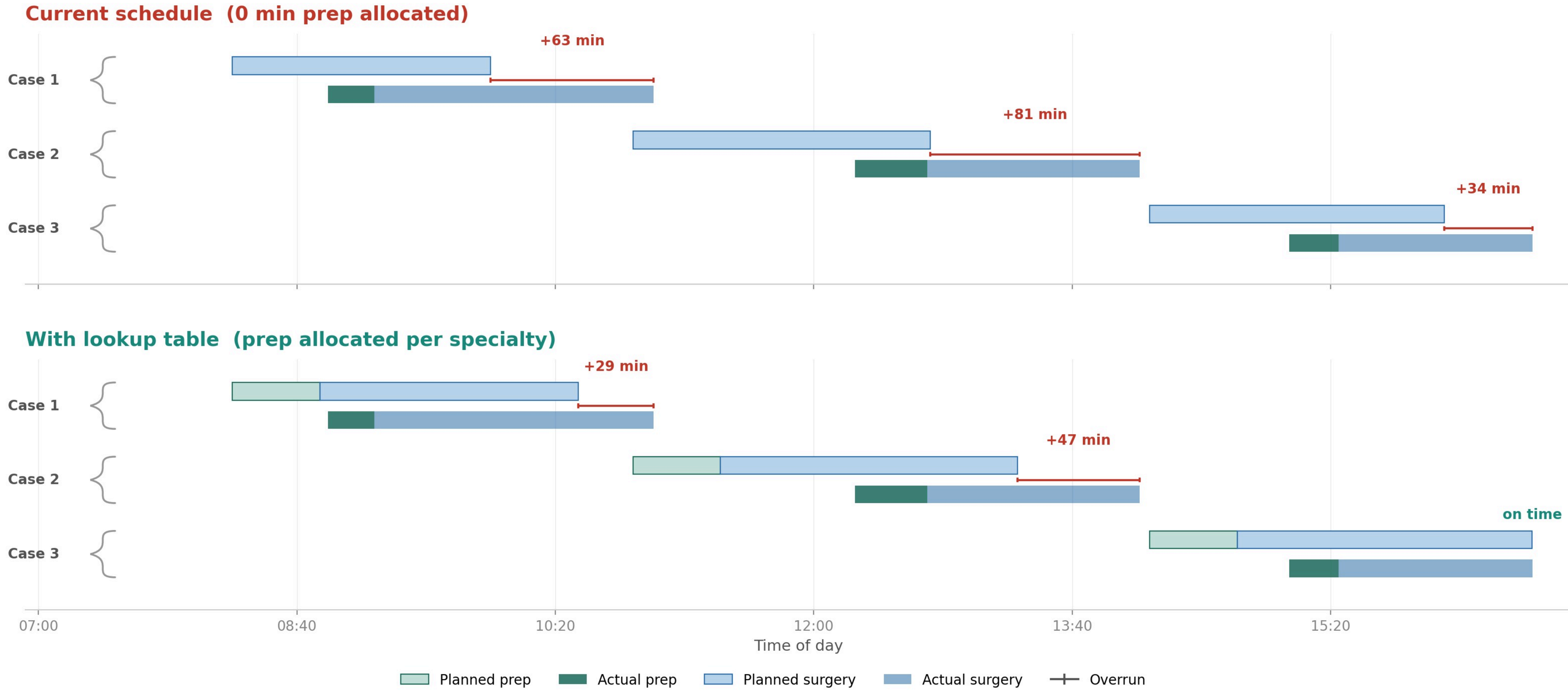
Solution: A 64-cell lookup table predicts what the schedule ignores


Mean
Median
Patient prep time (minutes)
140
120
100
80
60
40
20
0
n = 44,929
n = 75,033
No anesthesia
With anesthesia


R²
1.0
0.8
0.6
0.4
0.2
0.0
0.576
0.699
Lookup table
ML model


No anesthesia
With anesthesia
Arthroplasty 12 80
Anesthesiology 32 40
Breast surgery 44 65
Pediatric surgery 34 33
Pediatric orthopedics 16 53
⋮ 22 more ⋮
Trauma orthopedics 28 59
Tumor orthopedics 24 65
Urology 28 49
Ophthalmology 8 32
ENT 21 37
Mean prep time (min)
80
70
60
50
40
30
20
10

Impact: Same day, different schedule


Current schedule (0 min prep allocated)
Case 1
+63 min
Case 2
+81 min
Case 3
+34 min
With lookup table (prep allocated per specialty)
Case 1
+29 min
Case 2
+47 min
Case 3
on time
07:00
08:40
10:20
12:00
13:40
15:20
Time of day
Planned prep
Actual prep
Planned surgery
Actual surgery
Overrun

Mean overrun: +9.5 min → -19.5 min · Overruns reduced: 55% → 38% · ~20,600 fewer overruns

Note on data quality: 63.5% of room prep timestamps are recorded identically (start = end) — effectively unusable.

AI tools were used for coding assistance and ideation. All analysis, conclusions, and recommendations are the authors' own.